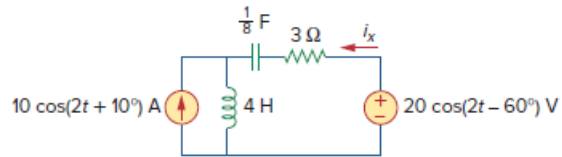


Problem

Using the superposition principle, find i_x in the circuit of Fig. 10.88.

Figure 10.88:



Step-by-step solution

Step 1 of 8

Refer to Figure 10.88 in the textbook.

There are two sources with same frequency; one way to obtain a solution is to use superposition theorem.

So let,

$$i_x = i_1 + i_2 \dots\dots (1)$$

Where,

i_1 is due to the $10 \cos(2t + 10^\circ) \text{ A}$ current source,

i_2 is due to the $20 \cos(2t - 60^\circ) \text{ V}$ voltage source.

Step 2 of 8

To find i_1 , consider the current source $10\cos(2t+10^\circ)\text{A}$ alone and short circuiting the voltage source $20\cos(2t-60^\circ)\text{V}$.

Determine the value of inductive reactance, X_L .

$$\begin{aligned} X_L &= j\omega L \\ &= j(2)(4) \\ &= j8\ \Omega \end{aligned}$$

Determine the value of capacitive reactance, X_C .

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2)\left(\frac{1}{8}\right)} \\ &= -j4\ \Omega \end{aligned}$$

Re-draw the circuit diagram.

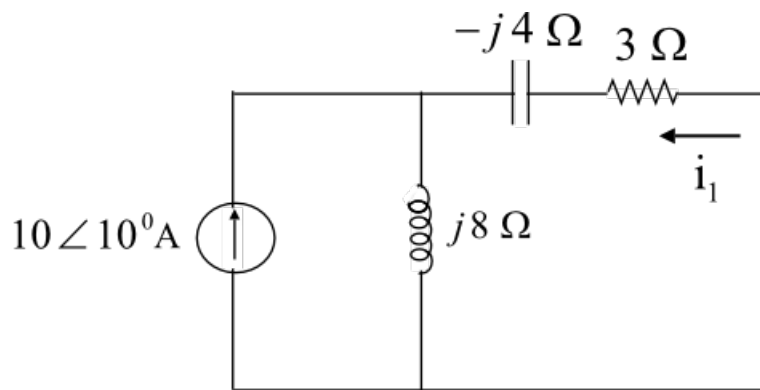


Figure 1

Step 3 of 8

Apply current division rule and determine the current, i_1 .

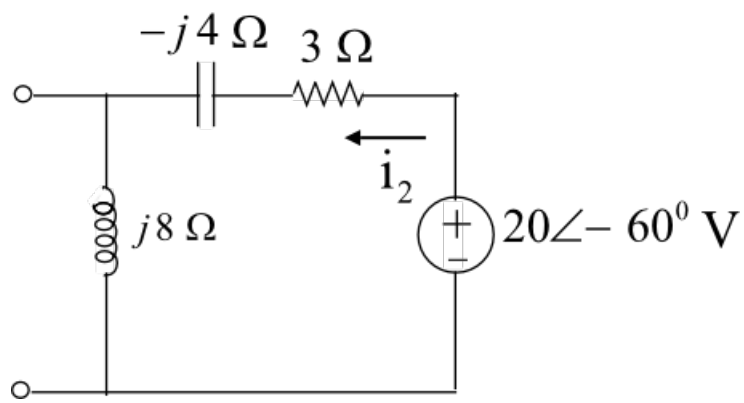
$$\begin{aligned} -i_1 &= 10\angle 10^\circ \times \frac{j8}{3-j4+j8} \\ i_1 &= -10\angle 10^\circ \times \frac{j8}{3+j4} \\ &= 10\angle (10^\circ+180^\circ) \times \frac{8\angle 90^\circ}{5\angle 53.13^\circ} \\ &= 10\angle (190^\circ) \times \frac{8\angle 90^\circ}{5\angle 53.13^\circ} \\ i_1 &= 16\angle -133.13^\circ\text{ A} \dots\dots (2) \end{aligned}$$

Step 4 of 8

To find i_2 , consider the voltage source $20\cos(2t-60^\circ)\text{V}$ alone and open circuiting the current source $10\cos(2t+10^\circ)\text{A}$.

Re-draw the circuit diagram.

Step 5 of 8



Step 6 of 8

Figure 2

Step 7 of 8

Apply ohm's law, determine the value of current, i_2 .

$$\begin{aligned} i_2 &= \frac{20\angle -60^\circ}{3 - j4 + j8} \\ &= \frac{20\angle -60^\circ}{3 + j4} \\ &= \frac{20\angle -60^\circ}{5\angle 53.13^\circ} \end{aligned}$$

$$i_2 = 4\angle -113.13^\circ \text{ A} \dots\dots (3)$$

Step 8 of 8

Substitute equation (2) and equation (3) in equation (1).

$$\begin{aligned} i_x &= 16\angle -133.13^\circ + 4\angle -113.13^\circ \\ &= 19.806\angle -129.17^\circ \text{ A} \end{aligned}$$

Therefore, the value of i_x is $19.806\angle -129.17^\circ \text{ A}$.